

Gabriel Kaszewski

EDUCATION

University of Gdańsk — B.S. in Bioinformatics

2021 – 2025

- Thesis: “*Modeling evolutionary processes using cellular automata*”
 - Built an evolutionary simulation in Rust using the Bevy engine (Entity Component System architecture); modeled predator-prey dynamics, mutation rates, and environmental pressures on cellular automata grids
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PUBLICATIONS

T. Urbańczyk, J. Koperski, **G. Kaszewski**, M. Synak, J. Swenda, M. Krośnicki.

“*Genetic algorithm for obtaining potential energy curve of diatomic molecules based on dispersed fluorescence spectra.*”

Scientific Reports (Nature Portfolio), 13, 17413 (2023).

DOI: 10.1038/s41598-023-44488-7

- Implemented the genetic algorithm optimization pipeline in Python (NumPy, threading, subprocess orchestration)
 - Software enabled novel method for reconstructing potential energy curves from experimental spectral data
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RESEARCH INTERESTS

Decentralized and federated systems — protocol design, availability analysis, and content distribution in peer-to-peer networks. Specifically: extending the ActivityPub protocol (W3C) for domain-specific content types, and studying availability degradation in federated networks under heterogeneous replication policies.

Proposed master’s thesis: “*Design and Implementation of a Decentralized Music Streaming Platform Using ActivityPub and Peer-to-Peer Content Distribution*” — federated music platform combining ActivityPub for social discovery with WebTorrent/IPFS for content delivery; availability evaluation under node churn and varying replication strategies.

SELECTED PROJECTS

k-ap — Generic ActivityPub Protocol Library Rust

[source](#)

- Reusable Rust crate implementing the ActivityPub server-to-server protocol: actor management, inbox/outbox routing, HTTP signatures, WebFinger, NodeInfo
- Published on a private Cargo registry; consumed as a dependency by two independent projects
- Designed around trait-based ports — consumers implement 7 repository traits to wire their own persistence and content model

Thoughts — Federated Microblogging Server Rust, Next.js, PostgreSQL, NATS

[source](#)

- Full ActivityPub-compatible microblogging platform (interoperates with Mastodon, Misskey, Pleroma)
- Hexagonal architecture with domain-driven design; async event fan-out via NATS JetStream
- Federation moderation: domain blocking, per-actor blocking with activity delivery

Movies Diary — Federated Movie Tracking Platform Rust, React, SQLite/PostgreSQL

[source](#)

- Self-hosted movie diary with ActivityPub federation, Jellyfin/Plex auto-import, full-text search, annual wrap-ups
 - 400+ unit tests; hexagonal architecture shared with Thoughts via the k-ap library
 - Dual interface: server-rendered HTML (zero JS) + React SPA (TanStack Router, shadcn/ui)
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INDUSTRY EXPERIENCE

Software Engineer — WPP (WPP Media / Choreograph / Wavemaker) Sep 2023 – Present

- Backend services (Python, FastAPI, Django) and frontend architecture (Angular, Module Federation) for an ad-tech platform serving 20,000 monthly users across a 9-person team
- Event-driven architecture on GCP: Pub/Sub, Vertex AI, Kubernetes; CI/CD via GitLab Pipelines

Software Engineer — GIAP Sp. z o.o. May 2021 – Feb 2023

- Built GeoGdańsk (geogdansk.pl), a public WCAG-compliant geospatial web application for the City of Gdańsk (React, TypeScript, ArcGIS JS API)
- Optimized PostGIS/PostgreSQL cross-database queries from 5+ minutes to under 15 seconds using CTEs

TECHNICAL SKILLS

Languages	Rust, Python, TypeScript, C#, C++
Frameworks	Axum, FastAPI, Django, React, Angular, Next.js
Infrastructure	Docker, Kubernetes, GCP, Azure, PostgreSQL, NATS, GitLab CI
Languages	Polish (native), English (C1)